

# HAML: Hierarchical Attractor Machine Learning with Bidirectional Coupling for Supervised Classification

A Preliminary Investigation

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## Abstract

We introduce HAML (Hierarchical Attractor Machine Learning), an exploratory supervised classification framework in which classification emerges from the convergence of a coupled dissipative dynamical system toward stable basins of attraction across multiple hierarchical levels. Each level operates in a PCA-decomposed subspace of decreasing dimension; levels are coupled through bottom-up (error-prediction) and top-down (context-prediction) feedback flows inspired by Predictive Coding.

**Primary contributions.** The paper makes two contributions we consider robust, and several theoretical observations that we present with explicit caveats:

- **Architectural** (Sections 2–3): a specific assembly of hierarchical PCA subspaces, gradient-learned continuous Gaussian attractors, and bidirectional PC-style coupling, trained end-to-end with a three-phase progressive schedule. The combination is original (Section 1.2) even though each individual primitive has direct antecedents.
- **Practical (Condition I1)** (Section 2.5): a dimension-scaled kernel initialisation  $\sigma_{\text{init}}^{(l)} = \sqrt{d_l} \cdot \sigma_{\text{data}}^{(l)}$  that prevents Gaussian-kernel saturation in high dimension and is empirically *necessary* for the framework to train beyond  $\sim 20$  dimensions (Table 2). Without I1, gradient magnitudes collapse to zero and learning halts.

**Theoretical observations (with caveats).** We present four theoretical observations that, on close inspection, are either specialisations of standard results or carry significant qualifications. We state these caveats upfront because they affect how the results should be read:

- Proposition 4.1 (dissipative convergence via LaSalle) is a standard property of any damped gradient flow with a bounded-below energy; in HAML, it justifies the ODE stopping criterion rather than constituting a deep result.
- Proposition 4.6 (dimension-independent local rate at fixed points) is mathematically correct but stands in *tension* with Condition II, which itself scales  $\sigma^2 \propto d_l$ . The two together imply that the local rate scales as  $1/d_l$  in practice; the result describes a property of the linearised regime, not a global dimension-independence guarantee.
- Proposition 4.10 (exponential expressivity gain in attractor count) is a worst-case representational statement; our experiments all use  $L \in \{2, 3\}$ , so the exponential factor is not empirically exercised.
- Theorem 4.15 (MAP equivalence with hierarchical PC under M1–M3) requires  $\lambda = 0$ , i.e. *disabling the repulsive terms* that are otherwise central to HAML. In the operational regime ( $\lambda > 0$ ), the energy decomposes as  $\mathcal{E} = \mathcal{F}_{\text{PC}} + \lambda \mathcal{F}_{\text{discrim}}$ , which is a PC–inference / contrastive-discriminative hybrid. The exact MAP correspondence applies only to a reduced version of the model.

The principled lesson is that HAML’s theoretical scaffolding is a coherent *specialisation* of established frameworks (LaSalle stability, neural-ODE generalisation theory, PC–inference correspondences), not a foundational contribution. We restate this explicitly in Section 4.

**Empirical results (honest summary).** Six campaigns across synthetic and real datasets. We summarise each with its actual statistical status:

- On a MNIST subset (1500 train / 500 test, 5 epochs), gradient training raises accuracy from 45.6% (k-means-only init) to 75.8%. **Bidirectional coupling provides no measurable gain** (75.8% coupled vs. 75.8% independent). This is a direct empirical *counter-observation* to the thesis that coupling is universally beneficial.
- On alternating concentric rings (5 seeds), coupled HAML reaches  $71.20\% \pm 6.43$  vs.  $67.30\% \pm 4.37$  for the independent variant. The +3.90 pts mean difference is *not statistically significant* at  $n = 5$ , though the qualitative shift to ring-aligned decision boundaries is robust across all five seeds.
- On noisy moons (**single seed**), coupling helps at intermediate noise (+6.27 pts at noise 0.20). Multi-seed validation pending.
- On Fashion-MNIST (784D, 10 classes, single seed, 10 epochs), test accuracy reaches 80.1%, *below standard neural baselines*. The result demonstrates stable training in 784D (enabled by II), not competitive accuracy.
- A controlled internal reimplementations of SA-nODE is dominated by HAML across noise levels (single seed; not using the authors’ official code).

**What the empirical evidence actually supports.** (i) Condition II is necessary and sufficient for high-dimensional training. (ii) The framework trains stably to 784D with the three-phase schedule. (iii) Bidirectional coupling produces a qualitative geometric advantage on structurally non-convex problems (rings), but its quantitative accuracy benefit is regime-dependent and is *negligible* on near-convex problems (MNIST). (iv) The combinatorial novelty of the architecture is preserved; the foundational novelty of any individual building block is not.

**Status of this work.** This is an exploratory preprint from an independent researcher. The code and configuration files needed to reproduce every reported number are open-source. We welcome external replication, criticism, and refutation.

**Positioning.** HAML sits at the intersection of supervised attractor-based classification, hierarchical Predictive Coding, hierarchical associative memories, and implicit-depth/equilibrium models. We claim no individual-component novelty; the contribution is the specific instantiation. Detailed positioning is provided in Section 1.2.

**Keywords:** Supervised learning, attractor dynamics, hierarchical classification, neural ODE, predictive coding, energy-based models, dynamical systems.

## 1 Introduction

Supervised classification traditionally relies on static function approximation through neural networks or kernel methods. We explore an alternative perspective: *classification as dynamic convergence*. Rather than computing a forward mapping  $f : \mathcal{X} \rightarrow \mathcal{Y}$  in one pass, the model learns to position attractors (stable equilibrium points) in hierarchical latent subspaces so that unseen samples converge to basins corresponding to their true class under coupled gradient dynamics.

This approach builds on four established research strands:

- **Dynamical systems theory:** stable manifolds and attractor landscapes provide geometrically interpretable decision boundaries.
- **Predictive coding** [5, 14]: top-down predictions correct bottom-up prediction errors across a generative hierarchy.
- **Energy-based models:** classification emerges from minimisation of a learned potential energy.
- **Modern Hopfield networks** [15]: continuous attractors replace discrete states for greater expressivity.

## 1.1 Scope and Disclaimer

We state upfront what this paper does and does not claim, to set expectations correctly.

**Hierarchy of contributions.** The paper has two contributions we consider *robust*:

1. **Architectural** (Sections 2-3): the specific combination of hierarchical PCA-decomposed subspaces, gradient-learned continuous Gaussian attractors, and bidirectional Predictive- Coding-style coupling, together with the three-phase progressive training schedule. The individual building blocks have direct antecedents (Section 1.2); the assembly is what we contribute.
2. **Practical (Condition I1)** (Section 2.5): a dimension-scaled kernel initialisation that is empirically *necessary* for the framework to function beyond  $\sim 20$  dimensions. Without I1, Gaussian kernels saturate and gradients vanish.

We additionally present *theoretical observations* (Section 4) which we make four important admissions about:

1. Proposition 4.1 (LaSalle dissipative convergence) is a *standard* property of damped gradient flows on energies bounded below; in our framework, it serves to justify the ODE stopping criterion, not as a deep result.
2. Proposition 4.6 (dimension-independent local rate) is in explicit *tension* with our Condition I1: with  $\sigma^2 \propto d_l$ , the scalar  $\kappa^{(l)} = w/\sigma^2$  scales as  $1/d_l$ . The result characterises a property of the linearised regime around a fixed point, not a global dimension-independence guarantee.
3. Proposition 4.10 (exponential expressivity gain in attractor count) is a worst-case representational statement; all our experiments use  $L \in \{2, 3\}$ , so the exponential factor is not exercised.
4. Theorem 4.15 (MAP equivalence with hierarchical PC) requires  $\lambda = 0$ , i.e. *disabling* the repulsive component that is otherwise central to HAML. In the operational regime ( $\lambda > 0$ ), HAML implements a PC-inference / contrastive-discriminative hybrid, not exact MAP inference.

**What this paper does NOT claim.** (i) State-of-the-art results on any benchmark. (ii) A universal empirical advantage of bidirectional coupling: the evidence is *mixed*, including a direct counter-observation on MNIST where the coupling provides no measurable gain. (iii) Superiority over SA-nODE based on the authors' official code (we use a controlled internal reimplementation). (iv) Multi-seed certification on real high-dimensional data. (v) Foundational theoretical novelty: the theoretical components are specialisations of established results, not new results in themselves.

**What we hope this paper offers.** A coherent, fully reproducible exploration of a specific architectural assembly at the intersection of four mature research strands, with an honest accounting of which components empirically pull their weight (the hierarchical PCA decomposition and Condition I1 do; the bidirectional coupling does so only in some regimes), and a starting point for further investigation by others.

## 1.2 Related Work and Positioning

HAML sits at the intersection of four research strands. We position our contribution explicitly with respect to each, listing the closest prior work and the specific differences with our framework.

**(i) Attractor-based supervised classification.** The idea of classifying via convergence to a basin of attraction has a long history. Amit and Mascaró [1] described a recurrent perceptron network in which each class is a pre-learned attractor of the recurrent dynamics, with the visual stimulus driving the system toward the correct class subset. Dense Associative Memory [9] and Modern Hopfield Networks [15] re-energised this line with continuous attractors and exponential storage capacity. Most directly, SA-nODE [12] introduces a continuous-time supervised classifier with pre-planted stable attractors in a single ambient space, trained end-to-end as a Neural ODE [4]. HAML differs from these by (a) operating in a *hierarchy* of subspaces rather than a single ambient space, (b) learning *continuous* attractor positions, widths, and repulsion radii by gradient descent rather than planting them analytically, and (c) introducing explicit inter-level coupling.

**(ii) Hierarchical Predictive Coding.** Predictive Coding (PC) posits a hierarchical generative model with top-down predictions and bottom-up prediction errors [5, 14]. Spratling [20] demonstrated that hierarchical PC can perform visual object recognition; Han et al. [7] introduced a deep bidirectional PC network with local recurrent processing whose internal representations *converge over time* during classification on SVHN, CIFAR, and ImageNet — structurally the closest prior work to HAML’s bidirectional coupling. The mathematical relationship between PC and backpropagation has been studied in depth [17, 19, 22], with results showing that PC networks can approximate or exactly replicate backpropagation under specific conditions. HAML differs from this line by (a) framing the dynamics as a continuous-time dissipative system with explicit attractor structure rather than as a discrete recurrent network, (b) using PCA-based subspace decomposition between levels, and (c) reading out classification directly from basin membership rather than from top-layer activity.

**(iii) Hierarchical associative memories.** Krotov [10] introduced a fully recurrent associative memory model with an arbitrary number of layers and a Lyapunov energy function that decreases along the dynamics. Salvatori et al. [16] realised associative memories via a hierarchical generative network

trained by predictive coding, with retrieval framed as inference. Both are very close to HAML’s hierarchical energy/attractor framing but target auto-associative memory and retrieval rather than supervised classification with class-specific basins.

**(iv) Implicit-depth and equilibrium models.** Deep Equilibrium Models (DEQ) [2] compute outputs as fixed points of an implicit transformation, with  $O(1)$  memory via implicit differentiation; Multiscale Deep Equilibrium Models (MDEQ) [3] extend this to hierarchical multi-resolution feature maps and achieve competitive results on ImageNet and Cityscapes. Monotone Operator Equilibrium Networks (monDEQ) [23] provide existence and uniqueness guarantees under monotonicity constraints. Generalisation bounds for neural ODEs have been derived via the Chen–Fliess series [8] and via path-norm arguments [13]. These works share with HAML the use of an implicit/continuous dynamics whose endpoint defines the output, but do not use a hierarchical attractor structure for class decision or a Predictive-Coding-style bidirectional coupling.

**Where HAML sits.** We do not claim that any individual building block (attractors, hierarchical PC, energy minimisation, fixed-point computation, dimension- scaled kernels) is new. To our knowledge, we did not find a published model that combines simultaneously: (a) a supervised-classification objective with read-out by basin membership, (b) a hierarchy of subspaces obtained by PCA decomposition, (c) continuous attractors learned by gradient descent at every level, and (d) a bidirectional PC-style coupling between adjacent levels. HAML’s contribution is this specific instantiation, together with the theoretical results that specialise to it. We treat the closeness of each building block to prior work explicitly when discussing limitations (Section 7.2).

Table 1 summarises the architectural differences with SA-nODE, the closest prior work in the *supervised-attractor-classification-via-ODE* family.

| Dimension             | SA-nODE                        | HAML                                         |
|-----------------------|--------------------------------|----------------------------------------------|
| State space           | Full input dim.                | Hierarchy of decreasing subspaces            |
| Attractors            | Binary eigenvectors $\pm a$    | Continuous, gradient-optimised               |
| Potential             | Fixed double-well              | Parametric Gaussian mixture (with repulsion) |
| Structure             | Single dynamic level           | $L$ bidirectionally coupled levels           |
| Coupling              | None                           | Bottom-up + top-down                         |
| High-dim. training    | Dimension-dependent saturation | Stabilised by Condition I1 (Sec. 2.5)        |
| Theoretical anchor    | Statistical mechanics          | Dyn. systems + Predictive Coding             |
| PC/MAP correspondence | Not established                | Holds for $\lambda=0$ subset (Thm. 4.15)     |

Table 1: Architectural comparison with SA-nODE [12]. We do not claim “dimension-independent convergence” in this row, in line with the caveats on Proposition 4.6 (see Section 4.2).

### 1.3 Contributions

The contributions of this paper, in decreasing order of robustness:

1. **Architectural assembly** (Section 2): the specific combination of (a) PCA-decomposed hierarchical state space, (b) gradient-learned continuous Gaussian attractors with explicit repulsion at every level, and (c) bidirectional PC-style inter-level coupling, trained end-to-end with a three-phase progressive schedule. The combination is not present in the prior work we reviewed.
2. **Condition I1 (dimension-scaled kernel initialisation)** (Section 2.5): an empirically *necessary* initialisation rule  $\sigma_{\text{init}} \propto \sqrt{d_l}$  that prevents Gaussian-kernel saturation in high dimension. Without I1, gradient magnitudes vanish and HAML cannot train beyond  $\sim 20$  dimensions (Table 2). This is the contribution that most directly enables the framework to function on real high-dimensional data.
3. **Practical training mechanisms**: the three-phase progressive coupling schedule (Section 3.1) and the heuristic level-aware collapse-recovery mechanism (Section 3.3).
4. **Preliminary empirical study**: 6 reproducible campaigns across synthetic and real datasets, with explicit reporting of single-seed limitations and missing baselines.
5. **Theoretical observations** (Section 4): four results — dissipative convergence, local rate at fixed points, hierarchical expressivity, MAP correspondence — presented with the caveats stated in Section 1.1 and explicit relations to prior work.
6. **Open reproducibility**: full source code at <https://github.com/kobootloader/HIERARCHICAL-ATTRACTOR-MACHINE-LEARNING>.

## 2 Model Architecture and Dynamics

### 2.1 Hierarchical State Space

Let  $\mathcal{X} \subset \mathbb{R}^d$  be the input space. We define a tower of linear projections:

$$\mathcal{X} = \mathcal{H}_0 \xrightarrow{\pi_1} \mathcal{H}_1 \xrightarrow{\pi_2} \dots \xrightarrow{\pi_L} \mathcal{H}_L \quad (1)$$

where  $\mathcal{H}_l \subset \mathbb{R}^{d_l}$  with  $d_0 > d_1 > \dots > d_L$ .

Projections  $\pi_l : \mathcal{H}_{l-1} \rightarrow \mathcal{H}_l$  are initialised by PCA (capturing maximum variance) and are optionally fine-tuned during training. Their Moore–Penrose pseudo-inverses  $\pi_l^\dagger : \mathcal{H}_l \rightarrow \mathcal{H}_{l-1}$  serve as upward decoders.

**Remark 2.1** (Contraction property). *PCA projections normalised to unit spectral norm satisfy  $\|\Pi_l\|_2 \leq \sqrt{d_{l+1}/d_l} < 1$ . This contraction is used in the convergence analysis (Section 4.1).*

Initial conditions:  $\mathbf{x}^{(l)}(0) = \pi_l \circ \dots \circ \pi_1(\mathbf{x}_{\text{input}})$ .

## 2.2 Attractors and Energy Potential

At level  $l$ , we maintain  $K \times M$  learnable attractors  $\{\boldsymbol{\mu}_{c,m}^{(l)}\}_{c \in [K], m \in [M]}$  with associated parameters  $(\sigma_{c,m}^{(l)}, \rho_{c,m}^{(l)}, w_{c,m}^{(l)})$ . The energy potential is a parametric attractor-repulsor mixture:

$$E^{(l)}(\mathbf{x}) = - \sum_{c=1}^K \sum_{m=1}^M w_{c,m}^{(l)} \exp\left(-\frac{\|\mathbf{x} - \boldsymbol{\mu}_{c,m}^{(l)}\|^2}{2(\sigma_{c,m}^{(l)})^2}\right) + \lambda \sum_{c \neq c'} \sum_{m,m'} w_{c',m'}^{(l)} \exp\left(-\frac{\|\mathbf{x} - \boldsymbol{\mu}_{c',m'}^{(l)}\|^2}{2(\rho_{c',m'}^{(l)})^2}\right) \quad (2)$$

**Bayesian interpretation** (anticipating Section 4.4): without the repulsive terms ( $\lambda = 0$ ),  $-E^{(l)}$  is proportional to the log of a Gaussian mixture, i.e. the log-prior  $\log p(\mathbf{x}^{(l)})$  in the implicit generative model. Repulsive terms correspond to a discriminative contrastive prior [11].

## 2.3 Coupled Dynamics

At each level  $l \in \{0, \dots, L\}$ , the state evolves under the coupled ODE:

$$\dot{\mathbf{x}}^{(l)}(t) = \mathbf{F}_{\text{intra}}^{(l)}(\mathbf{x}^{(l)}) + \mathbf{F}_{\text{bu}}^{(l)}(\mathbf{x}^{(l)}, \mathbf{x}^{(l-1)}) + \mathbf{F}_{\text{td}}^{(l)}(\mathbf{x}^{(l)}, \mathbf{x}^{(l+1)}) - \gamma \mathbf{x}^{(l)} \quad (3)$$

where  $\gamma > 0$  is a viscous damping coefficient ensuring energy dissipation.

**Intra-level force (local attractors and repulsors):**

$$\mathbf{F}_{\text{intra}}^{(l)} = -\nabla_{\mathbf{x}} E^{(l)}(\mathbf{x}^{(l)}) \quad (4)$$

**Bottom-up error-correction force:**

$$\mathbf{F}_{\text{bu}}^{(l)} = \alpha_{\text{bu}} \cdot \pi_l(\mathbf{x}^{(l-1)} - \pi_l^\dagger(\mathbf{x}^{(l)})) \quad (5)$$

This is the *prediction error* signal of Predictive Coding [14]: the difference between the actual lower-level state and the reconstruction predicted by the current level.

**Top-down prediction force:**

$$\mathbf{F}_{\text{td}}^{(l)} = \alpha_{\text{td}} \cdot \left(\pi_l^\dagger(\mathbf{x}^{(l+1)}) - \mathbf{x}^{(l)}\right) \quad (6)$$

This is the recall force toward the representation predicted by the higher level — the *top-down generation* in the Predictive Coding sense.

**Boundary conditions:**  $\mathbf{F}_{\text{bu}}^{(0)} = 0$ ,  $\mathbf{F}_{\text{td}}^{(L)} = 0$ .

## 2.4 Prediction

After integration until steady state (criterion:  $\|\dot{\mathbf{x}}^{(l)}\| < \epsilon \forall l$ ):

$$\hat{c} = \arg \max_c \sum_{l=0}^L \beta_l \sum_{m=1}^M \exp \left( -\frac{\|\mathbf{x}^{(l)}(T) - \boldsymbol{\mu}_{c,m}^{(l)}\|^2}{2(\sigma_{c,m}^{(l)})^2} \right) \quad (7)$$

where  $\beta_l$  are per-level weights (default:  $\beta_l = 2^l$ , learned or fixed).

## 2.5 Critical Design: Dimension-Dependent Kernel Initialisation (Condition I1)

**Problem:** in high dimension  $d$ , Euclidean distances concentrate around  $\mathbb{E}[\|\mathbf{x} - \mathbf{y}\|] \approx \sqrt{d} \cdot \sigma_{\text{data}}$ . With a fixed  $\sigma_{\text{kernel}} \sim 1.0$ , Gaussian kernels saturate toward 0 for  $d \gtrsim 20$ , forcing  $\nabla E \rightarrow 0$  and halting learning.

**Solution (I1):** initialise with dimension scaling:

$$\sigma_{\text{init}}^{(l)} = \sqrt{d_l} \cdot \sigma_{\text{data}}^{(l)} \quad (8)$$

**Empirical impact:**

| Dimension | Pre-fix kernel | Post-fix kernel | Gradient magnitude |
|-----------|----------------|-----------------|--------------------|
| 10        | 0.95           | 0.88            | 0.24               |
| 100       | 0.03           | 0.67            | 0.18               |
| 784       | < 0.01         | 0.52            | 0.15               |

Table 2: Kernel values and gradient magnitude by dimension (typical single run). Without I1, the framework fails to train beyond  $\sim 20$ D in our experiments.

## 3 Learning via Gradient Descent

### 3.1 Three-Phase Progressive Training

To stabilise learning of coupled dynamics, we use a three-phase schedule:

**Phase 1 (epochs  $1-E_1$ ): Independent learning.** Set  $\alpha_{\text{bu}} = \alpha_{\text{td}} = 0$ . Each level self-organises independently. Ensures conditions C1 and C3 (see Section 4.1) are empirically satisfied before coupling is introduced.

**Phase 2 (epochs  $E_1 + 1-E_1 + E_2$ ): Progressive coupling.**

$$\alpha_{\text{bu}}(t) = \alpha_{\text{bu}}^{\text{max}} \cdot \frac{t - E_1}{E_2}, \quad \alpha_{\text{td}}(t) = \alpha_{\text{td}}^{\text{max}} \cdot \frac{t - E_1}{E_2} \quad (9)$$

**Phase 3 (epochs  $E_1 + E_2 + 1-E_{\text{total}}$ ): Full coupling.** Fix  $\alpha_{\text{bu}} = \alpha_{\text{bu}}^{\text{max}}$ ,  $\alpha_{\text{td}} = \alpha_{\text{td}}^{\text{max}}$ .

### 3.2 Composite Loss Function

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \mu_1 \mathcal{L}_{\text{sep}} + \mu_2 \mathcal{L}_{\text{dyn}} \quad (10)$$

**Cross-entropy term** (multi-level softmax):

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N} \sum_{i=1}^N \log P(y_i \mid \{\mathbf{x}_i^{(l)}(T)\}_l) \quad (11)$$

**Basin separation term:**

$$\mathcal{L}_{\text{sep}} = \sum_l \sum_{c \neq c'} \sum_{m, m'} \exp \left( -\frac{\|\boldsymbol{\mu}_{c,m}^{(l)} - \boldsymbol{\mu}_{c',m'}^{(l)}\|^2}{2\sigma_{\text{min}}^2} \right) \quad (12)$$

**Dynamical regularisation:**

$$\mathcal{L}_{\text{dyn}} = \frac{1}{T} \int_0^T \sum_l \|\dot{\mathbf{x}}^{(l)}(t)\|^2 dt \quad (13)$$

Gradients are computed via RK4 unrolling (for short  $T$ ) or the adjoint method [4] via `torchdiffeq` (for long  $T$ , memory-efficient).

Attractor positions  $\boldsymbol{\mu}_{c,m}^{(l)}$  are initialised by per-class  $k$ -means on the projected training data, providing an initial configuration that empirically satisfies the local-dominance assumption (C1) on simple problems.

### 3.3 Level-Aware Collapse Recovery

**Proposition 3.1** (Level-Aware Recovery, informal mechanism). *Define a level  $l^*$  as “collapsed” when its single-level accuracy  $|\text{acc}_{l^*} - 1/K| \leq \varepsilon$  for at least *patience* consecutive epochs during Phase 3, while global train accuracy is moderate (anti-false-positive filter). When detected, the following is triggered:*

1. *Re-anchor attractors of level  $l^*$  via supervised prototype initialisation in its latent space.*
2. *Temporarily attenuate top-down forces with cooldown ( $\alpha_{\text{td}}$  scaled by  $\tau < 1$  for `td_cooldown_epochs`).*

This is a heuristic mechanism, not a theorem. Empirically on the concentric-rings benchmark (Campaign E8, 5 seeds 42–46), it provides a mean accuracy gain of +2.2 pts and a worst-seed gain of +4 pts. We treat this as engineering scaffolding, not as a contribution at the same epistemic level as the four theoretical propositions.

## 4 Theoretical Analysis

### Preamble: status of the theoretical results

Before presenting the results, we restate the epistemic position laid out in Section 1.1. Each of the four theoretical observations below *specialises* an established framework to HAML’s architecture; none constitutes a new theoretical result in its own right. We provide them because they clarify how HAML’s design choices relate to standard tools (LaSalle stability, Hessian analysis, topological complexity, hierarchical PC inference), and because the conditions under which each holds reveal genuine structural features of the framework. They should not be read as foundational contributions.

The four observations are summarised in Table 3 with their caveats made explicit. We recommend the reader consult this table while reading the formal statements below.

| Result                         | Substance                                                                                         | Caveat                                                                                                                                                                                    |
|--------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Prop. (LaSalle)                | 4.1 Damped gradient flow on bounded-below energy converges to critical points                     | Standard property of <i>any</i> such flow; serves as well-posedness of the ODE stopping criterion.                                                                                        |
| Prop. 4.3 (C1–C3)              | Correct-basin convergence under (C1) initial dominance, (C2) sufficient coupling, (C3) separation | Conditional and partly tautological; C2 is set by grid search, not verified analytically.                                                                                                 |
| Prop. 4.6 (local rate)         | Intra-level Hessian at a fixed point is $\kappa^{(l)}\mathbf{I}_{d_l}$ , scalar in $d_l$          | In <i>tension</i> with Condition I1: $\sigma^2 \propto d_l$ implies $\kappa \propto 1/d_l$ . The result describes the linearised regime only.                                             |
| Prop. 4.10 (expressivity)      | $O(M^L)$ representable configurations with $O(ML)$ parameters                                     | Worst-case representational; all experiments use $L \in \{2, 3\}$ , so the $M^L$ factor is not empirically exercised.                                                                     |
| Prop. (Rademacher)             | 4.13 Generalisation bound with sub-linear depth dependence                                        | Specialisation of existing neural-ODE bounds [8, 13]; not used operationally.                                                                                                             |
| Thm. 4.15 (MAP correspondence) | Energy equals hierarchical PC free energy under M1–M3                                             | M3 requires $\lambda = 0$ , which <i>disables</i> the repulsive terms central to HAML’s full model. The operational regime ( $\lambda > 0$ ) is a PC + contrastive hybrid, not exact MAP. |

Table 3: Theoretical results and their caveats. The detailed statements and remarks follow in this section.

### 4.1 Convergence Guarantees

In the overdamped regime (large  $\gamma$ , negligible inertia), the system reduces to a pure gradient flow:

$$\dot{\mathbf{X}} = -\frac{1}{\gamma}\nabla_{\mathbf{X}}\mathcal{E}(\mathbf{X}) \quad (14)$$

where  $\mathbf{X} = (\mathbf{x}^{(0)}, \dots, \mathbf{x}^{(L)})$  and the total energy is:

$$\mathcal{E}(\mathbf{X}) = \sum_l E^{(l)}(\mathbf{x}^{(l)}) + \frac{\alpha_{\text{bu}} + \alpha_{\text{td}}}{2} \sum_l \|\mathbf{x}^{(l)} - \pi_l^\dagger(\mathbf{x}^{(l+1)})\|^2 \quad (15)$$

**Proposition 4.1** (Dissipative Convergence).  *$\mathcal{E}$  is bounded below by  $-\sum_l \sum_{c,m} w_{c,m}^{(l)}$ . Along any trajectory,  $\dot{\mathcal{E}} = -\frac{1}{\gamma} \|\nabla \mathcal{E}\|^2 \leq 0$ . By LaSalle’s invariance principle, every bounded trajectory converges to the largest invariant subset of  $\{\mathbf{X} : \nabla \mathcal{E}(\mathbf{X}) = 0\}$ . The system can neither diverge nor oscillate indefinitely.*

**Remark 4.2** (Honest framing of Proposition 4.1). *This proposition is a standard property of damped gradient flows on energies bounded below; it is not specific to HAML. We state it because the integration code relies on it (the ODE solver terminates on the criterion  $\|\dot{\mathbf{x}}^{(l)}\| < \epsilon$ , which is well-defined precisely because of dissipative convergence). It does not guarantee convergence to a minimum, let alone to the correct-class minimum. Proposition 4.3 below formalises sufficient conditions for the latter, and is, we acknowledge, largely tautological in flavour.*

**Proposition 4.3** (Sufficient conditions for correct-basin convergence). *Assume:*

- C1 – Local dominance** : *at each level, the true-class attractor is the nearest attractor to the trajectory’s neighbourhood in Euclidean distance.*
- C2 – Sufficient coupling** :  $\alpha_{\text{bu}} + \alpha_{\text{td}} > \alpha_{\text{min}}$ , *a problem-dependent threshold determined by spectral analysis of the linearised system around fixed points.*
- C3 – Basin separation** :  $\mathcal{L}_{\text{sep}}$  *enforces a positive margin between attractors of different classes.*

*Then any trajectory whose initial condition lies in the joint true-class basin converges to the true-class fixed point at each level.*

**Remark 4.4** (Honest framing). *Proposition 4.3 is essentially: if the initial condition lies in the correct basin (C1) and the basins are separated (C3) and the coupling is well-tuned (C2), then gradient flow converges in that basin. We state it explicitly because (a) C1–C3 are precisely the conditions we monitor and enforce via  $k$ -means initialisation,  $\mathcal{L}_{\text{sep}}$ , and the three-phase schedule, and (b) the architectural value of HAML lies in making C1 hold at higher levels even when it fails at level 0, via top-down guidance (Proposition 4.5). The result is not deep in itself, but it makes the role of each design choice explicit.*

**Proposition 4.5** (Hierarchical Guidance). *Under coupling, the linearised dynamics around a fixed point have a block-tridiagonal Hessian (Eq. 16) whose*

minimum eigenvalue is bounded below by  $\min_l \lambda_{\min}(\mathbf{H}^{(l)}) + (\alpha_{\text{bu}} + \alpha_{\text{td}}) - 2\alpha_{\text{td}} \max_l \|\Pi_l\|_2$ . For PCA projections with  $\|\Pi_l\|_2 \leq 1$  and moderate coupling, this lower bound is strictly positive, ensuring exponential local convergence at every level once the trajectory enters the basin of the higher level.

The Hessian of  $\mathcal{E}$  at a fixed point has block-tridiagonal structure:

$$\mathbf{H} = \begin{pmatrix} \mathbf{H}^{(0)} + \mathbf{C}^{(0)} & -\alpha_{\text{td}} \Pi_1^T & 0 & \cdots \\ -\alpha_{\text{td}} \Pi_1 & \mathbf{H}^{(1)} + \mathbf{C}^{(1)} & -\alpha_{\text{td}} \Pi_2^T & \cdots \\ 0 & -\alpha_{\text{td}} \Pi_2 & \ddots & \vdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \quad (16)$$

with  $\mathbf{C}^{(l)} = (\alpha_{\text{bu}} + \alpha_{\text{td}}) \mathbf{I}_{d_l}$ . By Weyl’s interlacing inequality [21], the coupling matrices  $\mathbf{C}^{(l)}$  add a positive shift to the eigenvalues of the block-diagonal part; the off-diagonal  $-\alpha_{\text{td}} \Pi_l$  blocks act as a perturbation bounded by  $\alpha_{\text{td}} \|\Pi_l\|_2$ . In the moderate-coupling regime  $\alpha_{\text{td}} \|\Pi_l\|_2 < \alpha_{\text{bu}} + \alpha_{\text{td}}$ , the coupling strictly accelerates local convergence.

## 4.2 Local Hessian Structure at a Fixed Point

**Proposition 4.6** (Isotropic Local Hessian at a Fixed Point). *At a fixed point  $\mathbf{x}^{(l)*} = \boldsymbol{\mu}_{c,m}^{(l)}$ , the intra-level Hessian block of  $E^{(l)}$  in Eq. 2 reduces to  $\mathbf{H}^{(l)}|_{\boldsymbol{\mu}} = \kappa^{(l)} \cdot \mathbf{I}_{d_l}$ , where*

$$\kappa^{(l)} = \frac{w_{c,m}^{(l)}}{(\sigma_{c,m}^{(l)})^2} - \lambda \sum_{c' \neq c, m'} \frac{w_{c',m'}^{(l)}}{(\rho_{c',m'}^{(l)})^2} \exp\left(-\frac{\|\boldsymbol{\mu}_{c,m}^{(l)} - \boldsymbol{\mu}_{c',m'}^{(l)}\|^2}{2(\rho_{c',m'}^{(l)})^2}\right)$$

is a scalar parameterising the local relaxation rate  $T \sim \gamma/\kappa^{(l)}$ . The Hessian is isotropic across the  $d_l$  coordinates of the level.

**Remark 4.7** (Tension with Condition I1, and what the result actually says). *Earlier drafts of this preprint described this result as showing dimension-independent convergence. We retract that framing. The Hessian shape is isotropic in  $d_l$ , but its scalar magnitude  $\kappa^{(l)} = w/\sigma^2$  depends critically on how  $\sigma$  is set, and the two natural regimes have opposite issues:*

- Fixed  $\sigma$  (no dimension scaling):  $\kappa^{(l)}$  is independent of  $d_l$ , but the kernels saturate toward zero away from the fixed point (Table 2), so the dynamics never enter the linear regime: HAML fails to train at  $d_l \gtrsim 20$ .
- Condition I1 active ( $\sigma^2 \propto d_l$ ): kernels stay active globally and gradients are well-conditioned, but  $\kappa^{(l)} \propto 1/d_l$ , so the local relaxation time scales as  $T \sim \gamma d_l/w$ , growing linearly in  $d_l$ .

The genuine content of Proposition 4.6 is therefore: the Hessian’s isotropy means the linearised dynamics decouple uniformly across coordinates of a level, which is a structural property of Gaussian potentials absent from polynomial double-wells (SA-nODE). It is not a global “dimension-independence” guarantee. Condition I1 trades a  $d_l$  inflation of the local relaxation time against the ability to train at all in high dimension; this trade-off is what makes HAML operational at  $784D$ , not a free lunch.

**Corollary 4.8** (Depth and local rate). *Under moderate coupling ( $\alpha_{\text{td}} \max_l \|\Pi_l\|_2$  bounded uniformly in  $L$ ),  $\lambda_{\min}(\mathbf{H})$  of the coupled system remains bounded below by a quantity independent of  $L$  in the linear regime. Adding levels does not, by itself, slow local relaxation; the global convergence time and the dimension scaling through I1 are separate questions.*

### 4.3 Expressivity of the Hierarchy

**Definition 4.9** (Topological complexity). *The topological complexity  $\kappa(\mathcal{P})$  of a binary classification problem  $\mathcal{P}$  is the minimum number of connected components of the boundary of a Bayes-optimal decision region.*

**Proposition 4.10** (Hierarchical Expressivity Gain). *A flat ( $L = 1$ ) attractor model with one Gaussian attractor per simply-connected basin requires at least  $\kappa(\mathcal{P})$  attractors to represent a problem of topological complexity  $\kappa(\mathcal{P})$ . A hierarchical model with  $L$  levels and  $M$  attractors per level can represent up to  $O(M^L)$  distinct joint configurations  $(c, m_0, \dots, m_L)$  with only  $O(ML)$  parameters per class. On problems whose Bayes-optimal decision boundary factorises hierarchically (i.e. where a global discrete choice at level  $L$  selects a local geometry at level 0), the parameter count gap between flat and hierarchical is exponential in  $L$ .*

**Remark 4.11** (Caveats, including empirical scope). *Proposition 4.10 is an existence-style statement: it bounds the worst-case advantage of the hierarchy on a specific structural family of problems (hierarchically factorisable boundaries). It does not say that hierarchical models always beat flat ones — on problems without such factorisation, the gain may be zero.*

*A second caveat concerns the empirical scope of this paper. All our experiments use  $L \in \{2, 3\}$  (Sections 5.3–5.7), so the exponential factor  $M^L$  is never empirically exercised at scales where “exponential” is qualitatively distinguishable from “quadratic”. The representational gain we report is consistent with the proposition but does not demonstrate it. Validating the  $O(M^L)$  scaling experimentally would require campaigns with  $L \geq 4$  and comparison against  $L = 1$  baselines on problems of controlled topological complexity. We list this in Future Work.*

*A third caveat is on the decision rule. HAML’s prediction (Eq. 7) is a linear combination of per-class scores summed across levels, not a combinatorial function of the  $M^L$  joint configurations. The architecture represents*

$O(M^L)$  configurations in principle; the prediction rule does not exploit them combinatorially. A read-out that did exploit them is a possible extension.

The concentric-rings problem (Section 5.3) is a constructive example where the hierarchical factorisation holds: radial structure factorises out at level 1, angular refinement remains at level 0.

**Analytical example — alternating concentric rings:** classes defined by  $r \in [0, 1] \cup [2, 3]$  (class 1) and  $r \in [1, 2] \cup [3, 4]$  (class 2). With  $L = 1$  in the ambient 2D space, one needs  $\Omega(1/\sigma)$  attractors per ring to cover its angular extent. With  $L = 2$ , level 1 (radial projection,  $d_1 = 1$ ) resolves the global annular structure with 2 attractors per class; the top-down signal then constrains level 0 to refine locally.

**Remark 4.12** (Open empirical question: is bidirectional coupling necessary?). *The original version of this preprint stated this as a conjecture: that there exist classification problems representable by a hierarchical bidirectional system that are not representable by independent levels with the same attractor count. We now demote it to an open empirical question, because our own experiments are mixed:*

- *On the concentric-rings benchmark (Section 5.3), the coupled variant produces qualitatively different (ring-aligned, non-convex) decision boundaries that the independent variant does not, suggesting the coupling gives access to a representational regime independent levels do not reach.*
- *On the MNIST subset (Section 5.2), the coupled and independent variants achieve identical test accuracy (75.8% vs. 75.8%). This is a direct counter-observation: at least for this problem and this protocol, bidirectional coupling does not contribute representational power that independent levels lack.*

*The honest reading is that bidirectional coupling appears to matter for problems with genuine global-to-local hierarchical structure (rings) but not for problems that can be solved by local feature aggregation (MNIST at this scale). Establishing the boundary between these regimes empirically, and proving the corresponding separation result formally, is left to future work.*

**Proposition 4.13** (Rademacher Complexity Upper Bound). *For the coupled  $L$ -level classifier defined by Eq. 3 with  $K$  classes,  $M$  attractors per level, and  $n$  training samples, an upper bound on the empirical Rademacher complexity of the induced function class  $\mathcal{F}_L$  takes the form:*

$$\mathcal{R}_n(\mathcal{F}_L) \leq c \sqrt{\frac{KML}{n}} \cdot \prod_{l=1}^L (1 + \alpha_l \|\Pi_l\|_F) \quad (17)$$

for a universal constant  $c$  depending on the Lipschitz constants of the integrated flow. The dependence on depth is  $O(\sqrt{L})$  inside the square root and at most polynomial via the product term (which is bounded since  $\|\Pi_l\|_F \leq \sqrt{d_l}$  and PCA spectral norm is bounded).

**Remark 4.14** (Relation to prior generalisation results for implicit/continuous models). *The existence of Rademacher complexity bounds for neural ODEs and implicit-depth networks is not new: Hanson and Raginsky [8] derive bounds via the Chen–Fliess series for neural ODEs, and Marion [13] provides generalisation bounds for neural ODEs and deep residual networks under various path-norm assumptions. The specific structure we exploit here is the hierarchical-product form of the bound: the  $O(\sqrt{ML})$  scaling in the number of attractors and depth is a direct consequence of the union-bound argument over levels, and the product  $\prod_l (1 + \alpha_l \|\Pi_l\|_F)$  reflects the Lipschitz contraction afforded by PCA projections (cf. Remark on contraction in Section 4.2). We do not claim that this bound is tighter than existing bounds applied to the same architecture; we claim that it is the natural specialisation of the general theory to HAML’s specific structure. Closing the gap with the tightest known bounds for related implicit models is open work.*

## 4.4 Predictive Coding Correspondence and Bayesian Grounding

### 4.4.1 The Predictive Coding Generative Model

Predictive Coding [5, 14] posits a hierarchical generative model:

$$p(\mathbf{x}^{(0:L)}) = p(\mathbf{x}^{(L)}) \prod_{l=0}^{L-1} p(\mathbf{x}^{(l)} \mid \mathbf{x}^{(l+1)}) \quad (18)$$

In the point-mean-field approximation, the variational free energy reduces to:

$$\mathcal{F}|_{\text{pt}} = \sum_l \frac{1}{2} \|\boldsymbol{\varepsilon}^{(l)}\|_{\Sigma^{(l)}}^2 - \log p(\mathbf{x}^{(L)}) \quad (19)$$

where  $\boldsymbol{\varepsilon}^{(l)} = \mathbf{x}^{(l)} - g^{(l+1)}(\mathbf{x}^{(l+1)})$  are prediction errors. The inference dynamics are:

$$\dot{\mathbf{x}}^{(l)} = -\frac{\partial \mathcal{F}}{\partial \mathbf{x}^{(l)}} = -\boldsymbol{\varepsilon}^{(l)} + \left( \frac{\partial g^{(l+1)}}{\partial \mathbf{x}^{(l)}} \right)^T \boldsymbol{\varepsilon}^{(l+1)} \quad (20)$$

This is the structural form of HAML’s bidirectional coupling.

#### 4.4.2 Term-by-Term Identification

With linear generators  $g^{(l)}(\mathbf{x}^{(l)}) = \Pi_l^\dagger \mathbf{x}^{(l)}$  and isotropic covariances  $\Sigma^{(l)} = \frac{1}{\alpha_{\text{bu}} + \alpha_{\text{td}}} \mathbf{I}$ :

$$\mathcal{F}|_{\text{linear}} = \frac{\alpha_{\text{bu}} + \alpha_{\text{td}}}{2} \sum_l \|\mathbf{x}^{(l)} - \Pi_l^\dagger \mathbf{x}^{(l+1)}\|^2 - \log p(\mathbf{x}^{(L)}) \quad (21)$$

The coupling terms are identical to those in  $\mathcal{E}$  (Eq. 15). Identifying  $E^{(l)}$  with  $-\log p(\mathbf{x}^{(l)})$ :

$$-E^{(l)}(\mathbf{x})|_{\lambda=0} = \log \sum_{c,m} w_{c,m}^{(l)} \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_{c,m}^{(l)}, (\sigma_{c,m}^{(l)})^2 \mathbf{I}) + \text{const} \quad (22)$$

**Theorem 4.15** (MAP Equivalence under M1–M3). *Under conditions:*

**M1 – Linear generators** :  $g^{(l)}(\mathbf{x}^{(l)}) = \Pi_l^\dagger \mathbf{x}^{(l)}$  (satisfied by construction in our architecture).

**M2 – Isotropic covariances** :  $\Sigma^{(l)} = \frac{1}{\alpha_{\text{bu}} + \alpha_{\text{td}}} \mathbf{I}$  (satisfied when  $\alpha_{\text{bu}}$  and  $\alpha_{\text{td}}$  are level-independent; relaxable to per-level scalings).

**M3 – No repulsion** :  $\lambda = 0$  in Eq. 2.

With these three conditions,  $\mathcal{E} \equiv \mathcal{F}|_{\text{pt}}$  as functionals of  $\mathbf{X}$  (up to an additive constant), and HAML’s gradient flow exactly performs MAP inference in a hierarchical generative model with Gaussian mixture prior at each level and Gaussian conditional likelihoods.

With  $\lambda > 0$ ,  $\mathcal{E} = \mathcal{F}_{\text{PC}} + \lambda \mathcal{F}_{\text{discrim}}$ , where  $\mathcal{F}_{\text{discrim}}$  corresponds to a contrastive discriminative prior [11]. This hybrid is a discriminative augmentation of standard PC inference.

**Remark 4.16** (Critical caveat: M3 is *not* satisfied in HAML’s operational regime). The MAP equivalence requires condition M3 ( $\lambda = 0$ ), i.e. disabling the repulsive terms in Eq. 2. The repulsive terms are essential to HAML’s basin-separation mechanism (Section 3,  $\mathcal{L}_{\text{sep}}$ ); disabling them removes a component that the experimental code uses throughout ( $\lambda > 0$  in all reported runs). The implication is uncomfortable but should be stated openly:

- **In the regime where Theorem 4.15 holds exactly** ( $\lambda = 0$ ), HAML reduces to hierarchical Predictive Coding inference with a Gaussian-mixture prior. This is a known construction; HAML adds nothing foundational here.
- **In the regime where HAML actually operates** ( $\lambda > 0$ ), Theorem 4.15 does not apply exactly. The energy is the sum of a PC free-energy term and a contrastive discriminative prior [11], and HAML implements a PC + contrastive hybrid, not exact MAP inference.

We retain the theorem because the decomposition  $\mathcal{E} = \mathcal{F}_{\text{PC}} + \lambda \mathcal{F}_{\text{discrim}}$  is a useful structural observation about HAML’s energy, and because it suggests the natural VAE-style extension described in Section 7.3. But we no longer claim the MAP correspondence as a foundational contribution of this work: it is a specialisation of established results to a reduced version of HAML, and the full HAML is a known type of hybrid in the energy-based modelling literature.

**Remark 4.17** (Relation to prior PC–inference and PC–backprop correspondences). *The correspondence between Predictive Coding and variational inference in hierarchical Gaussian generative models is well established: it traces back to Rao and Ballard [14] and was formalised in Friston’s free-energy framework [5]. The relationship between PC and backpropagation has been the subject of substantial recent work: Whittington and Bogacz [22] showed that PC networks approximate the error backpropagation algorithm under specific conditions; Song et al. [19] established exact equivalence to BP on multilayer perceptrons under stronger assumptions; Salvatori et al. [17] extended this to arbitrary computation graphs.*

*Theorem 4.15 does not claim novelty for the PC–MAP correspondence in general. The result is a specialisation of these established correspondences to HAML’s specific generative model: Gaussian-mixture priors at each level (rather than Gaussian priors), PCA-projection generators (rather than learned MLPs), and isotropic inverse covariances tied to the coupling coefficients. We make conditions M1–M3 explicit precisely to highlight that the correspondence is conditional on these specific structural choices, and that M3 in particular excludes HAML’s full operational regime (see Remark 4.16 above).*

The implicit generative model is:

$$p(\mathbf{x}^{(l)}) = \sum_{c,m} w_{c,m}^{(l)} \mathcal{N}(\mathbf{x}^{(l)} \mid \boldsymbol{\mu}_{c,m}^{(l)}, (\sigma_{c,m}^{(l)})^2 \mathbf{I}) \quad (23)$$

$$p(\mathbf{x}^{(l-1)} \mid \mathbf{x}^{(l)}) = \mathcal{N}\left(\mathbf{x}^{(l-1)} \mid \Pi_l^\dagger \mathbf{x}^{(l)}, \frac{1}{\alpha_{\text{bu}} + \alpha_{\text{td}}} \mathbf{I}\right) \quad (24)$$

This Bayesian grounding suggests a natural VAE extension, replacing the point-mean-field approximation with an ELBO objective (Section 7.3).

## 5 Empirical Validation

We conduct six campaigns across synthetic and real datasets. We report results with their statistical status (single-seed vs. multi-seed, with means  $\pm$  standard deviations where available) and resist drawing conclusions that are not supported by the sample sizes.

### 5.1 Missing Baselines: A Limitation Acknowledged Upfront

We acknowledge that the empirical study does *not* include strong neural baselines (MLP, CNN) trained under identical protocols. On the datasets we use, established baselines achieve approximately:

- **MNIST (full)**: logistic regression  $\sim 92\%$ , MLP  $\sim 98\%$ , CNN  $> 99\%$ .
- **MNIST subset** (1500 train / 500 test, 5 epochs): a 2-layer MLP typically reaches  $90\%+$  in seconds.
- **Fashion-MNIST (5000 train)**: logistic regression  $\sim 84\%$ , 2-layer MLP  $\sim 88\%$ , small CNN  $> 90\%$ .

Our 75.8% on MNIST-subset and 80.1% on Fashion-MNIST are therefore *below standard baselines*. We do not present HAML as a more accurate classifier on these datasets. The empirical claims of this preprint concern (i) the framework trains stably in high dimension, (ii) the coupling provides a measurable benefit on structurally non-convex problems (rings, intermediate-noise moons), and (iii) the attractor geometry is interpretable and inspectable. Reaching baseline-competitive accuracy is left as explicit future work.

### 5.2 Campaign A: MNIST Subset — Gradient Training Gains and Coupling Neutrality

**Protocol:** 1500 train / 500 test from MNIST, levels  $784 \rightarrow 392 \rightarrow 196$ , 2 attractors per class, 5 epochs, 3-phase training, seed 42.

| Condition                                                  | Test Accuracy | Phase Progress (train) |
|------------------------------------------------------------|---------------|------------------------|
| $k$ -means init. only (no gradient training)               | 45.6%         | —                      |
| End of Phase 1 (independent levels)                        | —             | 64.2%                  |
| End of Phase 2 (progressive coupling)                      | —             | 68.7%                  |
| End of Phase 3 (full coupling)                             | —             | 71.8%–77.8%            |
| Full training, independent variant ( $\alpha = 0$ )        | 75.8%         | —                      |
| Full training, coupled ( $\alpha_{bu} = \alpha_{td} = 1$ ) | 75.8%         | —                      |

Table 4: Campaign A. Gradient training raises accuracy from 45.6% ( $k$ -means initialisation alone) to 75.8% (+30.2 pts). On this short, near-convex protocol, **bidirectional coupling provides no measurable benefit** over the independent variant. Single seed; the +30.2 pts gain compares to an extremely weak baseline ( $k$ -means class assignment with no gradient learning) and should not be interpreted as evidence of competitiveness with standard MNIST classifiers (see Section 5.1).

**Interpretation.** This is the most important negative result of the study: on a nearly-convex 10-class problem, coupling is neutral. This is consistent

with the theory (Proposition 4.10 requires high topological complexity to predict an advantage). We treat this result as informative for scoping the regime in which coupling helps, not as evidence against the architecture.

### 5.3 Campaign B: Concentric Rings — Coupling on Non-Convex Geometry

**Protocol:** alternating concentric rings (topological complexity  $\kappa \geq 2$ ),  $n_{\text{runs}} = 5$ , seeds 42–46.

| Model                               | Accuracy (mean $\pm$ std) | Qualitative boundary            |
|-------------------------------------|---------------------------|---------------------------------|
| Independent levels ( $\alpha = 0$ ) | 67.30% $\pm$ 4.37         | Convex (fails annular topology) |
| Coupled (tuned)                     | 71.20% $\pm$ 6.43         | Non-convex (aligns rings)       |
| $\Delta$ (mean)                     | +3.90 pts                 | —                               |

Table 5: Campaign B. The mean gain (+3.90 pts) is smaller than the pooled standard deviation ( $\sigma_{\text{coupled}} = 6.43$ ), so a Welch’s two-sample  $t$ -test on  $n = 5$  seeds per group does not reject the null hypothesis of equal means at standard significance levels. However, the qualitative shift in the learned decision boundary (visible in the per-seed plots in the repository) is consistent across all five coupled runs.

**Honest reading.** The quantitative mean difference is not statistically significant at  $n = 5$  seeds. The qualitative geometric shift (convex decision boundary in the independent variant; ring-aligned non-convex boundary in the coupled variant) is robust across all five seeds and is the actual evidence in favour of the coupling on this benchmark. The high variance in the coupled condition motivates Campaign E and the level-aware recovery mechanism.

### 5.4 Campaign C: Noisy Moons — Noise Robustness Profile (Single Seed)

**Protocol:** `make_moons` with noise  $\sigma \in \{0.10, 0.20, 0.30, 0.40\}$ ,  $n = 3000$ , single seed (42).

**Falsifiable prediction.** If the bell-shaped profile is real and not a single-seed artefact, multi-seed runs (10+ seeds) should reproduce: (a) peak coupling gain in the intermediate-noise regime, (b) no significant gain at very low or very high noise. We state this as a falsifiable prediction the reader can test against the released code.

| Noise | HAML-Indep. | HAML-Coupled  | KNN    | SVM RBF |
|-------|-------------|---------------|--------|---------|
| 0.10  | 99.60%      | 99.60%        | 99.87% | 99.87%  |
| 0.20  | 89.33%      | <b>95.60%</b> | 96.53% | 97.20%  |
| 0.30  | 88.13%      | 90.53%        | 90.80% | 91.33%  |
| 0.40  | 84.67%      | 85.07%        | 83.33% | 87.33%  |

Table 6: Campaign C. **Single seed.** The coupling delta (HAML-Coupled – HAML-Indep.) is 0.00, 6.27, 2.40, 0.40 pts at noise 0.10, 0.20, 0.30, 0.40. The maximum at intermediate noise is consistent with the prediction that top-down context helps most when local signal is ambiguous but exploitable. Multi-seed replication is required before treating the bell-shaped profile as established.

### 5.5 Campaign D: HAML vs. a Controlled SA-nODE Reimplementation

**Important caveat.** The SA-nODE comparison uses a *controlled internal reimplementation* of the architecture described in [12], not the authors’ official code, which we did not run for this study. The reimplementation reproduces: single ambient space, binary attractors  $\pm a$ , analytic double-well potential, linear coupling layer trained jointly with the dynamics, grid search over  $(a, \Delta t, \gamma)$ . A reviewer is justified in regarding this as a weak baseline: a paper comparison should ultimately use the official code. We report the result as preliminary.

| Noise | HAML Coupled  | SA-nODE (reimpl.) | Delta      |
|-------|---------------|-------------------|------------|
| 0.10  | <b>99.60%</b> | 86.67%            | +12.93 pts |
| 0.20  | <b>95.60%</b> | 86.40%            | +9.20 pts  |
| 0.30  | <b>90.53%</b> | 85.87%            | +4.67 pts  |
| 0.40  | <b>85.07%</b> | 83.33%            | +1.73 pts  |

Table 7: Campaign D. **Internal SA-nODE reimplementation, single seed.** The hierarchical architecture dominates our reimplementation of the flat single-space model across noise levels. SA-nODE degrades more slowly in absolute terms (−3.33 pts vs. −14.53 pts for HAML across the noise range), which we do not currently explain. We flag this comparison as preliminary pending evaluation against the official SA-nODE code.

### 5.6 Campaign E: Stabilisation and Inter-Level Collapse

We iteratively develop the level-aware collapse guard (Section 3.3) over subtests E1–E10. Full per-subtest results are in Appendix A.

| Subtest                         | Mechanism            | Mean Acc.         | Min (worst seed) |
|---------------------------------|----------------------|-------------------|------------------|
| E8 baseline (coupled)           | None                 | 68.60% $\pm$ 7.54 | 56.67%           |
| E8 with level-aware guard       | Sec. 3.3             | 70.80% $\pm$ 5.54 | 60.67%           |
| E10 ( $n = 3200$ , seeds 42–46) | Cause-oriented patch | 78.75% $\pm$ 7.63 | 68.25%           |

Table 8: Campaign E summary. The level-aware guard reduces variance and improves the worst-seed performance on the concentric-rings protocol. Scaling sample size from 1200 to 3200 (E10) lifts the mean accuracy by  $\approx 8$  pts.

**Mechanistic finding:** collapsed levels show `separation_ratio`  $\approx 0.15$  (low) despite moderate compactness — suggesting a learning blockade rather than total geometric collapse. Re-anchoring of the affected level’s attractors resolves the blockade within 1–2 epochs.

**Cross-dataset validation** (mutex baseline, seeds 42–46,  $n = 3200$ ):

- `noisy_moons` (noise 0.30): 89.13%  $\pm$  1.73, min 86.25%.
- `make_classification` (4 classes): 74.08%  $\pm$  4.31, min 67.12%.

## 5.7 Campaign F: Fashion-MNIST — Stable Training in 784D (Single Seed)

**Protocol:** Fashion-MNIST (784D, 10 classes),  $n_{\text{train}} = 5000$ ,  $n_{\text{test}} = 1000$ , levels  $784 \rightarrow 392 \rightarrow 196$ , 10 epochs, phases (2, 3, 5), **single seed (42)**.

| Metric                       | Value                      |
|------------------------------|----------------------------|
| Test accuracy                | <b>80.1%</b>               |
| Train accuracy (final)       | 82.3%                      |
| End of Phase 1               | 75.36%                     |
| End of Phase 2               | 78.86%                     |
| End of Phase 3               | 82.26%                     |
| Level-aware guard events     | 0 (naturally stable run)   |
| Compute (Colab GPU)          | 6 979 s ( $\approx 2$ hrs) |
| Reproducibility (Kaggle GPU) | 80.1% / 6 222 s            |

Table 9: Campaign F. **Single seed.** The headline result is that the three-phase training progresses smoothly through all phases in 784D without collapse, not that 80.1% is a strong accuracy on Fashion-MNIST (it is not; see Section 5.1). Identical results across two GPU environments confirm deterministic behaviour.

**Compute caveat.** At  $\approx 2$  hrs per 10-epoch run, a multi-seed Fashion-MNIST campaign was beyond the compute budget available for this preprint.

Multi-seed validation is the most important empirical follow-up.

## 6 Mechanistic Diagnostics

We implement diagnostic tools to isolate failure modes:

### 6.1 Level-Wise Accuracy Tracking

At each epoch, per-level accuracy  $\text{acc}_l(t)$  is logged. Divergence  $\Delta_l = \max_l \text{acc}_l - \min_l \text{acc}_l$  serves as a collapse indicator. Typical seed-42 signature on the concentric-rings task:

- Phase 1 (independent): all levels converge to  $\sim 60\%$ .
- Phase 2 (progressive): divergence emerges ( $\Delta_l \approx 0.15$ ).
- Phase 3 (full coupling): one level plateaus at  $\sim 0.5$ ; guard triggers re-anchoring.

### 6.2 Attractor Geometry Metrics

Per level:

- **intra\_spread\_mean**: within-class attractor spread (compactness).
- **inter\_centroid\_dist\_min**: minimum between-class centroid distance.
- **separation\_ratio**: geometric margin proxy.

Collapsed levels show **separation\_ratio**  $\approx 0.15$  (low) despite non-zero compactness, indicating a learning blockade rather than total geometric collapse.

### 6.3 Force Distribution Analysis

Histograms of  $\|\mathbf{F}^{(l)}\|$  per level reveal:

- *Saturated kernels* (pre-fix I1 in high dimensions):  $\|\mathbf{F}\| \approx 0$ .
- *Balanced learning* (post-fix I1): approximately Gaussian distribution.
- *Gradient death* in late phases: indicator of convergence or instability.

## 7 Discussion

### 7.1 What the Evidence Supports, and What It Does Not

#### Evidence supports:

1. The framework trains stably in dimensions up to 784, given condition I1.
2. On structurally non-convex problems (concentric rings), coupled HAML *qualitatively* recovers ring-aligned decision boundaries that the independent variant fails to produce.
3. Bidirectional coupling provides a single-seed accuracy gain at intermediate noise on noisy moons (consistent with theory; awaiting multi-seed confirmation).
4. Exact MAP correspondence with hierarchical Predictive Coding under explicit conditions M1–M3 (as a specialisation of the general PC–inference correspondence; see Remark 4.17).

#### Evidence does not support:

1. HAML being competitive with standard neural baselines on MNIST or Fashion-MNIST on accuracy alone.
2. A statistically significant mean accuracy gain from coupling at  $n = 5$  seeds on rings.
3. Superiority over SA-nODE based on the authors’ official implementation.
4. Scalability beyond 784D.
5. Novelty of any individual building block taken in isolation.

### 7.2 Genuine Limitations

1. **Combinatorial rather than foundational novelty:** each architectural primitive of HAML — continuous attractors, hierarchical PC, energy minimisation, fixed-point computation, dimension-scaled kernels — has direct antecedents in the literature reviewed in Section 1.2. The contribution is the specific assembly of these primitives for supervised classification by basin membership, not the invention of any of them. Readers should evaluate HAML as a synthesis with motivated specialisation, not as a new family of models.
2. **Missing strong baselines:** no MLP, CNN, or boosted-tree baseline trained under identical protocols. This is the most important methodological gap.

3. **Single-seed runs on real data:** MNIST-subset and Fashion-MNIST are single-seed for compute reasons.
4. **Statistical underpowering on rings:**  $n = 5$  seeds is insufficient to claim a mean accuracy advantage from coupling; we rely on qualitative geometric evidence.
5. **Internal SA-nODE reimplementation:** not a substitute for evaluation against the authors' official code.
6. **Computational cost:**  $\approx 2$  hrs per 10-epoch run on Fashion-MNIST (5000 samples). Adjoint methods and gradient checkpointing should reduce this substantially.
7. **Inter-level instability:** variance remains high between seeds; the level-aware guard helps but does not fully resolve it.
8. **Hyperparameter sensitivity:**  $(\alpha_{bu}, \alpha_{td}, \mu_1, \mu_2)$  require per-dataset tuning. Automated search is an open engineering challenge.
9. **Unexplained asymmetry:** SA-nODE-reimpl. degrades more slowly than HAML as noise increases (Campaign D); we do not currently have a mechanistic explanation.
10. **Open empirical question on bidirectional coupling:** Remark 4.12 now records that bidirectional coupling appears beneficial on rings but not on MNIST. Establishing the structural boundary between these regimes is open.
11. **Theoretical specialisation, not foundational novelty:** the theoretical results in Section 4 are specialisations of existing frameworks (LaSalle stability for gradient flows; Rademacher complexity for neural ODEs [8, 13]; PC-inference correspondence [17, 19, 22]) to HAML's specific architecture, not foundational results in their own right.

### 7.3 Future Work

1. **Strong baselines:** MLP / CNN / boosted trees under identical protocols on all reported datasets.
2. **Multi-seed Fashion-MNIST:** at least 5 seeds with the adjoint method to amortise compute.
3. **Official SA-nODE comparison:** replication against Marino et al.'s released code.
4. **Adjoint integration:** full neural-ODE adjoint sensitivity to reduce memory and training time.

5. **Generative extension (HAML-VAE)**: replace the point-mean-field approximation with an ELBO objective.
6. **Resolution of Remark 4.12**: structural characterisation of problems where bidirectional coupling actually adds representational power.
7. **Tighter Rademacher constants** and sample complexity bounds.
8. **Scalability beyond 784D**: tested on CIFAR-10 (3072D) and larger.

## 8 Reproducibility and Implementation

All code is available at: <https://github.com/ikobootloader/HIERARCHICAL-ATTRACTOR-MACHINE-LEARNING>

### 8.1 Key Software

- `haml/`: main package (dynamics, model, training, projection, analysis, visualisation).
- `experiments/`: 6 reproducible campaign scripts (A–F).
- `diagnose_forces.py`: high-dimension saturation detector.
- Dependencies: `numpy`, `scikit-learn`, `torch`, `torchdiffeq`, `scipy`.

### 8.2 Reproducible Commands

**Campaign A** (MNIST ablation):

```
python experiments/mnist_coupling_ablation.py
```

**Campaign B** (concentric rings, multi-seed):

```
python experiments/concentric_coupling_ablation.py --n-runs 5 --save-figure
```

**Campaign C** (noisy benchmark):

```
python experiments/noisy_benchmark.py
```

**Campaign F** (Fashion-MNIST, GPU):

```
PYTHONIOENCODING=utf-8 python experiments/fashion_mnist_short_probe.py \
--device cuda --seed 42 --n-train 5000 --n-test 1000 --n-epochs 10 \
--phase1-epochs 2 --phase2-epochs 3
```

## 9 Conclusion

We introduced HAML, a supervised classification framework that combines hierarchical PCA-decomposed state spaces, gradient-learned continuous Gaussian attractors, and bidirectional Predictive-Coding-style coupling. The contributions, in decreasing order of robustness, are: (i) an original architectural assembly of these primitives; (ii) Condition I1, a dimension-scaled kernel initialisation that is empirically necessary for high-dimensional training; (iii) a practical three-phase progressive training schedule and a heuristic collapse-recovery mechanism; and (iv) a preliminary empirical study with explicit single-seed limitations.

The four theoretical observations in Section 4 (dissipative convergence, isotropic local Hessian, hierarchical expressivity, PC-MAP correspondence) are *specialisations* of established results to HAML’s specific architecture, with caveats stated explicitly in Section 1.1 and Table 3. They are not foundational theoretical contributions. In particular, Proposition 4.6 stands in tension with Condition I1 (the trade-off is laid out in Remark 4.7), and Theorem 4.15 applies exactly only when the repulsive component of HAML is disabled (Remark 4.16).

Empirically, the evidence is preliminary and mixed. The framework trains stably in 784D (80.1% Fashion-MNIST, single seed, below standard baselines). Bidirectional coupling provides a qualitative geometric advantage on structurally non-convex problems (rings) and a single-seed accuracy gain at intermediate noise (moons), but *no measurable gain* on near-convex problems (MNIST subset). The latter is a direct empirical counter-observation to the thesis that hierarchical coupling is universally beneficial, and we report it explicitly rather than hide it. Multi-seed validation and comparison against strong neural baselines remain the most important next steps.

We release the framework and all experimental code in the hope that it will be useful as a starting point for further investigation of attractor-based classification, hierarchical Predictive Coding implementations, and energy-based hybrids. We welcome external replication, criticism, and refutation.

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| Subtest | Key mechanism                                       | Accuracy (test)    | $\Delta$ vs. indep. |
|---------|-----------------------------------------------------|--------------------|---------------------|
| E1      | Reinforced coupling ( $\alpha_{td} = 1.5$ , 25 ep.) | 86.38%             | +15.63 pts          |
| E2      | Early-stop + LR decay                               | 82.88%             | +12.00 pts          |
| E3      | Adaptive $\mu_{sep}$ + soft landing ep. 20          | 69.88%             | −1.25 pts           |
| E4      | Soft landing on divergence trigger                  | 76.25%             | +6.37 pts           |
| E5      | Raised trigger + patience 2                         | 76.13%             | +5.00 pts           |
| E6      | Collapse guard on divergence jump                   | 86.38%             | +16.00 pts          |
| E7      | Seed-wise diagnostics + LR cascade patch            | $69.33\% \pm 7.10$ | +9.60 pts           |
| E8      | Level-aware guard (phase 3 only)                    | $70.80\% \pm 5.54$ | +11.07 pts          |
| E10     | Scale $n = 3200$ , cause-oriented patch             | $78.75\% \pm 7.63$ | —                   |

Table 10: Stabilisation iteration summary (Campaign E). Results on the concentric-rings protocol, except E10 where  $n = 3200$ , seeds 42–46. The high single-seed peaks (E1, E6) were not retained because their stability across seeds is not established. E8 was selected as the operational mechanism on the multi-seed mean–variance trade-off.

## A Detailed Campaign Results

### A.1 Campaign E — Subtest Summary

### A.2 Initialisation Condition I1 — Justification

Distance concentration in  $\mathbb{R}^d$ :

$$\mathbb{E}[\|\mathbf{x} - \mathbf{y}\|] \approx \sqrt{d} \cdot \sigma_{\text{data}} \quad (25)$$

For a Gaussian kernel  $\exp(-r^2/2\sigma^2)$  to maintain activations in the range  $[0.1, 0.9]$  across dimension  $d$ ,  $\sigma$  must scale as  $\sqrt{d}$ . Without I1, the following table shows the resulting kernel saturation (single typical run).

| Dimension | Pre-fix kernel | Post-fix kernel | Gradient |
|-----------|----------------|-----------------|----------|
| 10        | 0.95           | 0.88            | 0.24     |
| 100       | 0.03           | 0.67            | 0.18     |
| 784       | < 0.01         | 0.52            | 0.15     |

Table 11: Kernel activations and gradient magnitudes with and without condition I1.

## B Hyperparameter Sensitivity

Ablation on the concentric-rings protocol:

## C Summary of Theoretical Results

| Parameter            | Range tested | Accuracy range |
|----------------------|--------------|----------------|
| $\alpha_{\text{bu}}$ | [0.0, 1.0]   | [67.3%, 73.2%] |
| $\alpha_{\text{td}}$ | [0.0, 1.5]   | [67.3%, 75.1%] |
| $\mu_1$ (separation) | [0.1, 1.0]   | [70.5%, 71.9%] |
| $\mu_2$ (dynamics)   | [0.001, 0.1] | [71.1%, 71.3%] |

Table 12: Sensitivity on primary hyperparameters. The model is moderately robust;  $\alpha_{\text{bu}}$  and  $\alpha_{\text{td}}$  require the most attention per dataset.

| Result                                                       | Section     | Substance                                                         | Caveat / status                                                                                   |
|--------------------------------------------------------------|-------------|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Prop. (LaSalle)                                              | 4.1 4.1     | Damped gradient flow converges to critical points.                | Standard for any damped gradient flow on bounded energy.                                          |
| Prop. 4.3 (C1–C3)                                            | (C1– 4.1    | Correct-basin convergence under explicit conditions.              | C1–C3 are not verified analytically; partly tautological.                                         |
| Prop. 4.5 (Hierarchical guide)                               | (Hier- 4.1  | Coupling shifts Hessian eigenvalues positively.                   | Standard Weyl perturbation argument.                                                              |
| Prop. (Isotropic Hessian)                                    | 4.6 4.2     | Local Hessian at fixed point is $\kappa^{(l)} \mathbf{I}_{d_l}$ . | In <i>tension</i> with I1: $\kappa \propto 1/d_l$ with I1. Not a global dim-independence.         |
| Prop. 4.10 (Expressivity)                                    | (Ex- 4.3    | $O(M^L)$ configurations with $O(ML)$ parameters.                  | Worst-case; $L = 2, 3$ in all experiments.                                                        |
| Rem. 4.12 (Bidi. open Q)                                     | (Bidi. 4.3  | Bidirectional coupling helps on rings, neutral on MNIST.          | Empirically open.                                                                                 |
| Prop. (Rademacher)                                           | 4.13 4.3    | Sub-linear-in-depth complexity bound.                             | Specialisation of existing neural-ODE bounds; not used operationally.                             |
| Thm. (MAP)                                                   | 4.15 4.4    | $\mathcal{E} = \mathcal{F}_{\text{PC}}$ under M1–M3.              | M3 ( $\lambda = 0$ ) excludes HAML’s operational regime. Specialisation of known correspondences. |
| <i>Non-theoretical contributions (the more robust ones):</i> |             |                                                                   |                                                                                                   |
| Cond. I1                                                     | 2.5         | $\sigma_{\text{init}} \propto \sqrt{d_l}$ initialisation.         | Empirically necessary for $d > 20$ .                                                              |
| Prop. 3.1 (Level-aware recovery)                             | (Level- 3.3 | Heuristic to re-anchor collapsed levels.                          | Engineering, not a theorem.                                                                       |

Table 13: Summary of theoretical results, with their epistemic status. The genuinely robust contributions of the paper (architectural assembly, Condition I1) are listed at the bottom for comparison.